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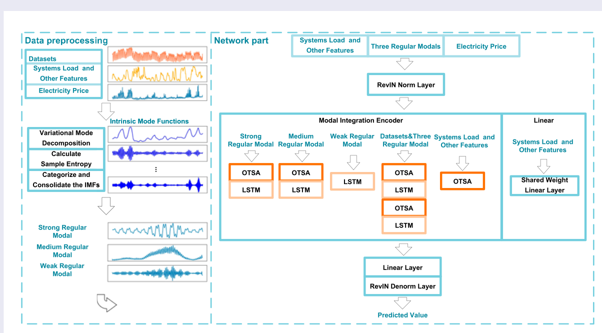
Lisen Zhao, Lihua Lu and Xiang Yu

School of Electronics and Information, Shanghai Dianji University, Shanghai, People's Republic of China

ABSTRACT

The electricity market is a complex and dynamic environment characterized by a multitude of factors that influence electricity prices. Accurate and reliable electricity price forecasting (EPF) is crucial for market participants, including power generators, consumers, and policymakers. Electricity prices are influenced by temporal dependencies and electricity consumption patterns. Therefore, dependencies across different feature dimensions (cross-dimensional dependencies) and temporal trend information are essential. To address the aforementioned issues, we propose Multimodal Integration and Linear Enhanced Transformer (MILET), which combines cross-dimensional dependencies with single-dimensional modal features. First, we decompose electricity price data into three regular modals using Variational Mode Decomposition and Sample Entropy. This approach enables us to uncover the intrinsic patterns within the variable, thereby simplifying the complexity of the data series. Then integrate these three modals and the original dataset into a five-channel encoder (Modal Integration Encoder, MIE) with both single and multi-dimensional information. MIE is composed of Overall Two-Stage Attention (OTSA) and Long Short-Term Memory (LSTM), where OTSA handles cross-dimensional dependencies, and LSTM addresses long-term dependencies. Additionally, we capture trend information in electricity consumption features through linear layers and linearly integrate the data to obtain the forecasting results. Extensive experimental results on five electricity price datasets demonstrate the effectiveness of MILET compared to state-of-the-art techniques. Our code is available at https://github.com/Lisen-Zhao/MILET/tree/master.



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Electricity price forecasting; Crossformer; two-stage attention; variational mode decomposition; sample entropy

1. Introduction

Electricity Price Forecasting (EPF) (Weron, 2014) is an important tool for improving economic benefits. Its application is crucial for achieving energy transition (Tschora et al., 2022). By leveraging price predictions to profit in the market, it promotes the development of smart applications such as optimizing electric vehicle batteries (Beltrán et al., 2022; Lago et al., 2018). As the penetration of renewable energy sources in the power system increases, solving the EPF problem becomes increasingly challenging. Price fluctuations are often influenced by factors such as supply and demand, time, electricity generation, electricity consumption, and the proportion of different electricity generation systems, resulting in sudden peaks in price fluctuations (Hosius et al., 2023; Sai et al., 2023). These characteristics further contribute to the instability and non-linearity of electricity prices. Experiments are often conducted to analyze the dependency between features and the target, such as multivariate forecasting. In such contexts, the challenge lies in the

CONTACT Lihua Lu lulihua@sdju.edu.cn

high volume of information contained within the predicted target sequence, making the extraction of relevant features from this sequence a crucial step in the prediction process. EPF encompasses three primary types of models to address this challenge: traditional statistical models (Che & Wang, 2010; Gürtler & Paulsen, 2018; Tan et al., 2010), machine learning models (Kapoor & Wichitaksorn, 2023; Yang et al., 2022), and deep learning models (Dalal et al., 2023; Gareta et al., 2006; Memarzadeh & Keynia, 2021), each bringing its own strengths and approaches to the task of feature extraction and prediction.

In traditional statistical methods, exponential smoothing (Jónsson et al., 2014) involves taking a weighted average of historical data to predict future time points. The weighting coefficients follow a geometric decay, with data closer in time periods being assigned greater weights, and the sum of these weights equals 1. This method addresses the issue of weight distribution for points at different distances, accommodating the challenges posed by both overall averaging and moving regression (Beigaite et al., 2018). AutoRegressive Moving Average (ARMA) (Dash & Dash, 2019) combines autoregressive (AR) and moving average (MA) components to capture the dependencies between past observations and the current value of a time series. It excels in modelling stationary sequences. AutoRegressive Integrated Moving Average (ARIMA) (Tan et al., 2010) adds differencing to the combination of AR and MA components, making it be capable of handling nonlinear and non-stationary time series. These statistical methods are simple and highly descriptive, but their accuracy decreases as the forecasting horizon extends, and they exhibit weaker performance in fitting highly volatile datasets and strongly nonlinear data.

Machine learning methods can capture complex nonlinear relationships. Support Vector Regression (SVR) (Zahid et al., 2019) primarily aims to find an optimal hyperplane that minimizes the error between actual and predicted values while allowing for a certain level of tolerance. SVR is effective in capturing complex relationships within data and proves useful for addressing nonlinear regression problems by mapping data to a higher-dimensional space using kernel functions. Random Forests (Wang et al., 2022), on the other hand, combine predictions from multiple decision trees to make more accurate and robust forecasts. By creating numerous decision trees during training and averaging their outputs for classification or regression tasks, this ensemble approach mitigates overfitting and enhances generalization by incorporating a diverse set of decision trees.

In the field of deep learning models, the currently predominant models are the Transformer (Vaswani et al., 2017) and its derivative (S. Liu et al., 2021), as well as hybrid neural network models (Y. Liu & Zou, 2022). Transformer models, which employ self-attention to capture long-term dependencies at different time steps in Long-term Time Series Forecasting (LTSF) (Li et al., 2019), have been highly successful and applied in various domains. In the time series domain, numerous variant models have been developed with the aim of reducing model complexity while maintaining or improving performance. Informer (H. Zhou et al., 2021) extends the Transformer model using ProbSparse attention based on KL-divergence. Autoformer (Wu et al., 2021) replaces the traditional self-attention mechanism with an Auto-Correlation module, achieving $O(LlogL)$ complexity. Reformer (Kitaev et al., 2020) examines how self-attention operations scale with sequence length and proposes an alternative attention mechanism where attention blocks no longer include separate projections for queries but still ensure performance. FEDformer (T. Zhou et al., 2022) reduces complexity to $O(L)$ by applying frequency domain self-attention. DLinear (Zeng et al., 2023) suggests that many previous Transformer models based on cross-time attention fail to adequately learn cross-time dependencies and proposes using linear models with shareable weights, which outperform some Transformer models. Crossformer (Y. Zhang & Yan, 2022) utilizes a Two-Stage Attention (TSA) layer, primarily designed to capture cross-temporal dependencies that result from segmenting embeddings by dimensions, as well as dependencies among different dimensions.

Hybrid neural network models (H. Liu et al., 2021) have shown good performance in improving prediction accuracy. Fang and He (2023) propose an improved predictive model that combines adaptive decomposition with Recurrent Recursive Networks (RNN) for multi-feature fusion. B. Shao et al. (2023) introduce a fusion forecasting method based on Variational Mode Decomposition (VMD), Sample Entropy (SE), Gate Recurrent Unit (GRU), and Support Vector Quantile Regression (SVR-QR). They classify decomposed curves by calculating entropy and accumulate high information change curves, achieving enhanced prediction accuracy and interval forecasting. Z. Shao et al. (2022) present a framework called MHSAC-NLSTM, which combines Multi-Head Self-Attention (MHSA), LSTM, and Convolutional Neural Networks (CNN) to enhance prediction efficiency. This framework leverages the synergy between these components to improve predictive accuracy and efficiency. Y. Zhang et al. (2023) propose a wastewater pipeline total phosphorus (TP) concentration prediction model that combines Empirical Mode Decomposition (EMD) with Ensemble Noise, SE, VMD, and LSTM. They perform a two-level decomposition of the original time series into curves with lower complexity. The experiments compare many approach to highlighting LSTM's nonlinear mapping capabilities.

Although many studies have mad contributions to EPF in performance and accuracy, it can be still improved. Our goal is to further optimize these aspects. The main contributions of this paper include:

(1) In terms of model architecture, we introduced the Overall Two-Stage Attention (OTSA) without dimension embedding cutting, which prevents information loss. Within the encoder section, we proposed the 5-channel Modal Integration Encoder (MIE) that combines single-dimensional modality and multidimensional data processing based on OTSA and modal decomposition. Using the encoder described above as the foundation, we developed the MILET model, which integrates linear layers for trend capturing.
(2) We validate our proposed model: MILET which demonstrates state-of-the-art performance in five datasets from different countries within the context of the electricity market. Comparative analyzes with previous state-of-the-art techniques highlight its superiority.
(3) Through various Ablation Experiments, they can verify the Efficiency and Emphasize Interpretability of our proposed model: MILET.

2. Our model: MILET

MILET (Multimodal Integration and Linear Enhanced Transformer) is specifically tailored for Electric Power Forecasting (EPF). It operates with a diverse set of inputs, encompassing historical price data and supplementary forecasting data, including electricity consumption, among other variables. The forecasting process can be described as Equation (1).

$$\mathbf{Y}_{l:l+n} = \psi\left(\mathbf{X}^{D}_{:l-1}\right) \tag{1}$$

Where the input denoted as $\mathbf{X} \in \mathbb{R}^{D}_{L}$, is a 2D array with a historical length of L and D dimensions. ψ represents the neural network model that processes this input. $\mathbf{Y}$ is the output, with a length of n, where the time positions are located after the initial l time steps.

EPF is inherently a problem of converting multidimensional inputs into a single-dimensional output. The crux of our model lies in its ability to consolidate multiple dimensions into a unified, single-dimensional output. While current deep learning models are primarily focused on cross-dimensional dependencies, with complex network models offering solutions for these, we must also consider the intrinsic patterns within each individual feature. Focusing exclusively on cross-dimensional dependencies does not adequately address the complete extraction of information from single features.

Current network models, such as those based on the Transformer architecture, can struggle with both overfitting or underfitting when dealing with single-dimensional feature extraction. They may not effectively capture latent features, and they often lack interpretability, making them susceptible to getting trapped in local optima. To address these challenges, we introduce MILET, a framework designed to effectively integrate single-dimensional and multidimensional forecasting, as depicted in in Figure 1.

2.1. MILET

Based on above approach, the process diagram of the MILET model is depicted as follows. Initially, we preprocess the data by decomposing the electricity price component of the dataset using VMD to obtain mulitple IMFs. Subsequently, we calculated the SE for each IMF. Based on the magnitude of entropy values, we partitioned and integrated the IMFs, ultimately categorizing them into three regular modals. The strong regular modal is composed of the IMFs with the highest entropy values. The IMFs with the lowest entropy are integrated into weak regular fluctuations, resembling residual components. Finally, the remaining IMFs are integrated to form moderate regular fluctuations, similar to trend components. These three modals are concatenated with the dataset to form the model's input. The structure diagram of the MILET model is depicted in Figure 2.

Within the network, we first perform a RevIn (Kim et al., 2021) layer on all data, which ensures that the data distribution remains unbiased between the test and training sets, thereby reducing errors. Considering the similarity between these three modals, separate structures are designed to fit each of them, extracting intrinsic features.

The network for data processing in this paper is divided into two parts: the linear part and the Modal Integration Encoder (MIE). The former primarily consists of a linear layer with shared weights, focusing on learning trends from the remaining features in the dataset. The latter is composed of OTSA and LSTM concatenated for each of the three modes based on their sample entropy. For strong regular components, we employ the higher-complexity OTSA-LSTM, while for weak regular components, we use only LSTM. In the MIE, we also consider cross-dimensional dependencies among all features, utilizing a dual-layer OTSA-LSTM for all features. On the remaining features, besides fitting trends using a linear layer, we also apply OTSA individually. Consequently, we

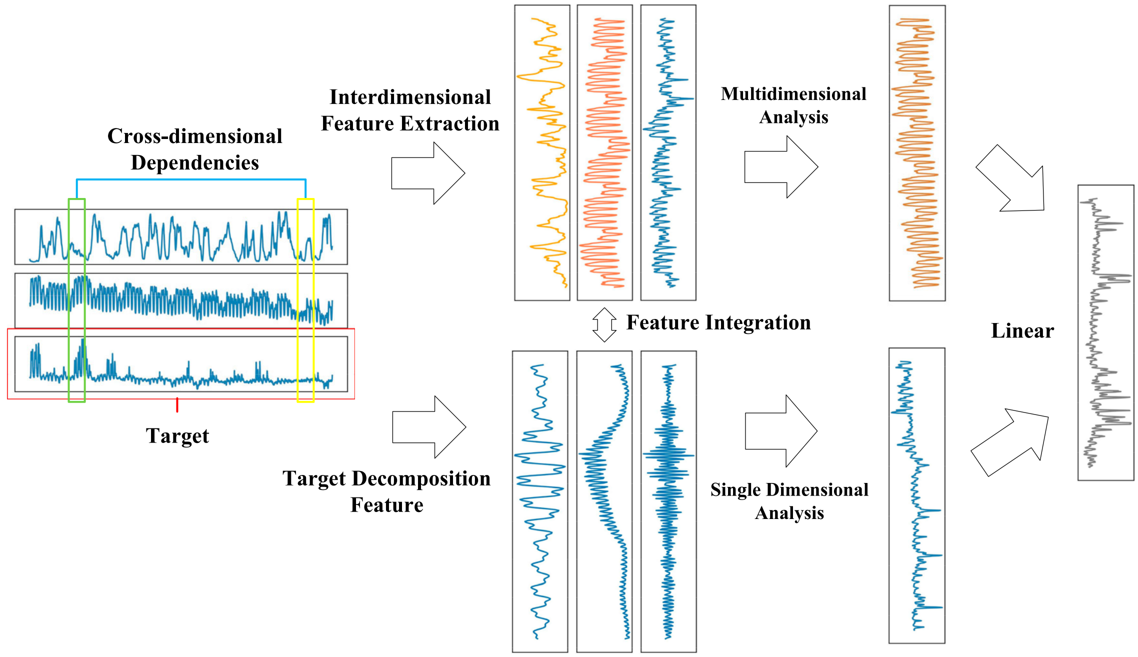


Figure 1. Feature extraction model architecture diagram based on cross-multi-dimensional and target dimension decomposition. The green and yellow boxes highlight cross-dimensional dependencies, while the red boxes signify target feature predictions. Following feature extraction and decomposition, a composite of these features is used for subsequent predictions. The results are then segregated for both multi-dimensional and single-dimensional analysis, culminating in the prediction outcome through linear layer integration.

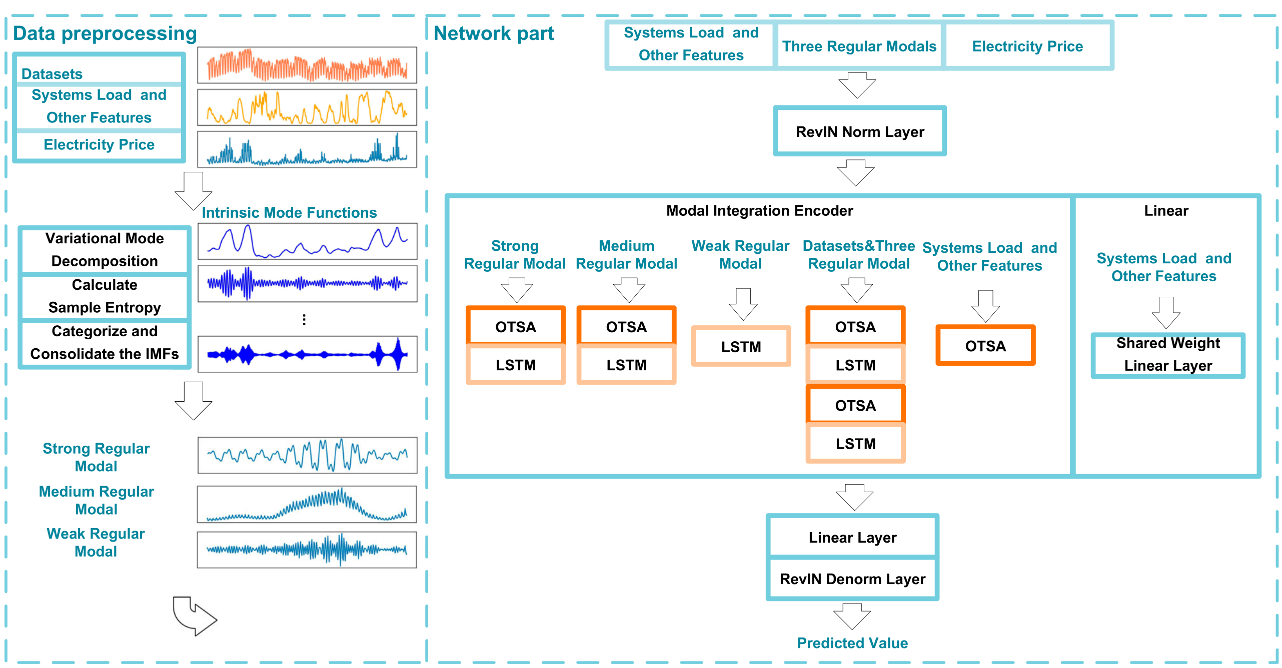


Figure 2. MILET model structure diagram. Data preprocessing generates modals from feature decompositions, and the Network part accepts both the dataset and modals. MIE (Modal Integration Encoder) predicts with varying complexities based on different data combinations. Shared Weight Linear Layer gathers overall trends, while a Linear Layer before RevIN Denorm Layer integrates output channels.

obtain five output channels within the MIE, which are linearly combined to produce the MIE's output. This output is integrated with the output from the linear part and passed through a RevIn Denorm layer to obtain the final output. In the integration process, Dropout (Srivastava et al., 2014) is applied at each layer.

2.2. Overall two-stage attention

The Overall Two-Stage Attention (OTSA): building upon the Two-Stage Attention (TSA) (Y. Zhang & Yan, 2022), sequentially extracts features across time and dimensions from the data, and its mechanism yields favourable predictive outcomes. However, during the attention mechanism after the segmentation of the embedding operation, the process of cross-transformation of the dataset can result in information loss (Gao et al., 2023). Additionally, the additional time embedding layer for encoding time series can lead to a decrease in model performance. Therefore, we propose the OTSA, which directly performs two-stage attention on the entire dataset.

2.2.1. Cross-time stage

For processing a 2D array, the matrix $\mathbf{S} \in \mathbb{R}^{D \times L}$, where $D \geq 1$ serves as the input of OTSA, D represents the dimensions, allowing for the case of 1, L represents the feature length. Unlike TSA, which segments the data and extracts time-length vectors, OTSA relies on MSA under the mode of $D = 1$ to capture high-weight time-dependent information. $\mathbf{S}$ is the output of the lower-level OTSA or input data containing electricity price data. The time-cross attention part is depicted in Equations (11) and (12).

$$\hat{\mathbf{S}}^{Time} = \mathrm{LayerNorm}(\mathbf{S} + \mathrm{MSA}(\mathbf{S}, \mathbf{S}, \mathbf{S})) \tag{2}$$

$$\mathbf{S}^{Time} = \mathrm{LayerNorm}\left(\hat{\mathbf{S}}^{Time} + \mathrm{MLP}\left(\hat{\mathbf{S}}^{Time}\right)\right) \tag{3}$$

Where $\hat{\mathbf{S}}^{Time}$ represents the output of the MSA applied to the time-crossed dimensions, $\mathbf{S}^{Time}$ is the output of time-cross attention itself. LayerNorm denotes Layer Normalization (Ba et al., 2016), with a computational complexity of $O(D^2L)$.

2.2.2. Cross-dimension stage

To reduce complexity in the cross-dimension stage, we introduce a routing mechanism, creating a router matrix $\mathbf{R} \in \mathbb{R}^{c \times L}$, $(c << D)$ as an index for retrieving dimensional features. Where c is the designed indexing variable, and its value is significantly smaller than D. Unlike TSA's multi-dimensional routing variables, we allow the existence of a single dimension in the previous stage. Therefore, c varies based on the input, reducing the complexity from $O(D^2L)$ to $O(DL)$. The dimension-cross attention part is illustrated by Equations (13), (14), (15) and (16).

$$\mathbf{B} = \mathrm{MSA}_1^{dim}\left(\mathbf{R}, \mathbf{S}^{Time}, \mathbf{S}^{Time}\right) \tag{4}$$

$$\bar{\mathbf{S}}^{dim} = \mathrm{MSA}_2^{dim}\left(\mathbf{S}^{Time}, \mathbf{B}, \mathbf{B}\right) \tag{5}$$

$$\hat{\mathbf{S}}^{dim} = \mathrm{LayerNorm}\left(\mathbf{S}^{Time} + \bar{\mathbf{S}}^{dim}\right) \tag{6}$$

$$\mathbf{S}^{OTSA} = \mathrm{LayerNorm}\left(\hat{\mathbf{S}}^{dim} + \mathrm{MLP}\left(\hat{\mathbf{S}}^{dim}\right)\right) \tag{7}$$

Where $\mathbf{R}$ represents the router, $\mathbf{B}$ encompasses all cross-dimensional information, $\bar{\mathbf{S}}^{dim}$ denotes the output of the routing mechanism, and there are skip connections between $\hat{\mathbf{S}}^{dim}$, $\mathbf{S}^{OTSA}$, $\mathbf{S}^{Time}$ and $\hat{\mathbf{S}}^{Time}$. This router query mechanism contributes to more efficient computations and enhances the search for temporal segment associations, ultimately improving the information available for the MSA step.

2.3. Variational mode decomposition (VMD)

VMD (J. Zhang et al., 2020) is a non-recursive signal processing technique aimed at minimizing the non-smoothness of complex and nonlinear time series data. It does this by iteratively finding the best variational modes and breaking them down into multiple components with restricted bandwidths (Huang et al., 2021). The variational construction expression is shown in Equation (2):

$$\min_{\{\mu_m\},\{\omega_m\}} \left\{ \sum_{m=1}^{M} \left\| \partial_t \left[(\delta(t) + j/\pi t) * \mu_m(t) \right] e^{-j\omega_m t} \right\|_2^2 \right\}$$
$$\text{s.t.} \sum_{m=1}^{M} \mu_m = f \tag{8}$$

The expression involves sub-sequences with central frequencies ω_m, Dirac distribution δ, and Lagrange multiplier $\lambda(t)$ with a quadratic penalty term α to ensure constraint integrity and reconstruction accuracy. The Lagrangian is detailed in Equation (3).

$$\begin{aligned} &L\left(\{\mu_m\}, \{\omega_m\}, \lambda\right) \\ &= \alpha \sum_{m=1}^{M} \left\| \partial_t \left[(\delta(t) + j/\pi t) * \mu_m(t) \right] e^{-j\omega_m t} \right\|_2^2 \\ &+ \left\| f(t) - \sum_{m=1}^{M} \mu_m(t) \right\|_2^2 \\ &+ \left\langle \lambda(t), f(t) - \sum_{m=1}^{M} \mu_m(t) \right\rangle \end{aligned} \tag{9}$$

An alternating direction method optimizes modal components and frequencies, solving a variational problem.

Minimization points are updated alternately, following Equations (4), (5), and (6).

$$\hat{\mu}_m^{n+1}(\omega) = \frac{\hat{f}(\omega) - \sum_{i \neq m} \hat{\mu}_i(\omega) + \hat{\lambda}(\omega)/2}{1 + 2\alpha\,(\omega - \omega_m)^2} \tag{10}$$

$$\omega_m^{n+1} = \frac{\int_0^\infty \omega \left|\hat{\mu}_m^{n+1}(\omega)\right|^2 d\omega}{\int_0^\infty \left|\hat{\mu}_m^{n+1}(\omega)\right|^2 d\omega} \tag{11}$$

$$\hat{\lambda}^{n+1}(\omega) = \hat{\lambda}^n(\omega) + \tau\left(\hat{f}(\omega) - \sum_{m=1}^{M} \hat{\mu}_m^{n+1}(\omega)\right) \tag{12}$$

Equations (3) and (4) apply Fourier transforms to $\mu(t)$, $\omega(t)$, and $\lambda(t)$, with nn as the iteration index and α as the penalty factor. Larger α values reduce the bandwidth of each IMF component.

2.4. Sample entropy

VMD modal functions preserve the core of original variables, and IMF analysis enhances clarity, sometimes at the cost of complexity (Qiao & Yang, 2020). Sample Entropy (SE) is utilized to evaluate the variability of information content across subsequences, with lower SE indicating greater regularity (Duan et al., 2021; Y. Zhang et al., 2016). We analyze sequences by generating m-dimensional vectors at discrete time points, each spanning m values from position i, to identify the maximum variation, as per Equation (7).

$$d\left[X_m(i), X_m(j)\right] = \max_{k \in [0, m-1]} |x(i+k) - x(j+k)| \tag{13}$$

For these values, a tolerance deviation denoted as r is introduced for measurement. The number of cases where $d[X_m(i), X_m(j)] < r$ is counted and denoted as B_i. The ratio of B_i to the total number of vectors and its average value are calculated using the Equations (8) and (9):

$$B_i^m(r) = \frac{B_i}{l - m - 1} \tag{14}$$

$$B^m(r) = \frac{1}{l - m} \sum_{i=1}^{N-m} B_i^m(r) \tag{15}$$

Expanding from m dimensions to $m+1$ dimensions, the process described above is repeated to obtain $B_{m+1}(r)$. Finally, SE is computed using the Equation (10).

$$\mathrm{SampEn}(m, r, l) = -\ln\left[\frac{B^{m+1}(r)}{B^m(r)}\right] \tag{16}$$

2.5. Long short-term memory

LSTM performs well in time series forecasting, effectively addressing long-term dependency issues in long time series problems (Yu et al., 2019). The features extracted by the OTSA mechanism, after passing the LSTM layer, can efficiently discard weak correlations and retain essential information needed for memory. The relevant equations of the LSTM are given as Equation (17).

$$\begin{aligned}
f_t &= \sigma\left(W_f \cdot \left[h_{t-1}, x_t\right] + b_f\right) \\
i_t &= \sigma\left(W_i \cdot \left[h_{t-1}, x_t\right] + b_i\right) \\
\tilde{C} &= \tanh\left(W_C \cdot \left[h_{t-1}, x_t\right] + b_C\right) \\
C_t &= f_t * C_{t-1} + i_t * \tilde{C}_t \\
o_t &= \sigma\left(W_o\left[h_{t-1}, x_t\right] + b_o\right) \\
h_t &= o_t * \tanh\left(C_t\right)
\end{aligned} \tag{17}$$

3. Experiments

3.1. Dataset

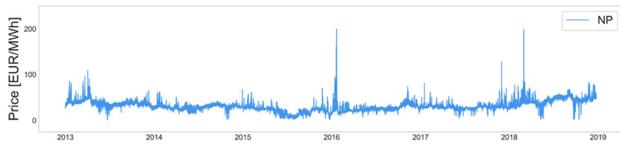
The experimental dataset in this paper consisting of five publicly available benchmark datasets from open-source python library epftoolbox (Lago et al., 2021), represents electricity markets in various regions. These datasets exhibit distinct frequency and volatility patterns in electricity price data, allowing for testing the model's generalization ability. They also have sufficiently long durations for the model to thoroughly analyze the data. The data covers relatively recent years, incorporating the impact of new energy sources and aligning with current technological developments.

Electricity price fluctuations depend on exogenous factors, so each dataset includes two aspects of electricity consumption data specific to each electricity system. The datasets have a length of 2184 days, equivalent to 6 years or 312 weeks. Missing data points in these datasets are interpolated between two available values. The following are detailed descriptions of the five datasets:

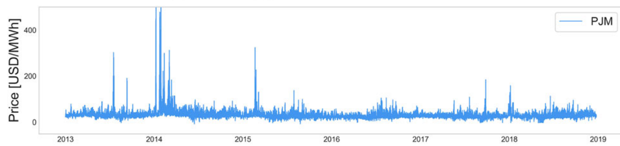
NP (Nordic Power): This dataset represents the Nordic power market and includes electricity price data, day-ahead load forecasts, and day-ahead wind power forecasts. The time span covers from January 1, 2013, to December 24, 2018.

PJM: The PJM dataset represents the Pennsylvania-New Jersey-Maryland (PJM) electricity market. It aligns with the NP dataset until December 24, 2018, and includes zonal prices from Commonwealth Edison (COMED), along with related system loads and COMED latitude load forecasts. The visualization of the NP and PJM datasets is presented in Figure 3.

EPEX-BE (Electricity Exchange Belgium): This dataset represents the electricity market in Belgium. It includes two additional dimensions: day-ahead load forecasts for the previous day and day-ahead wind power generation forecasts for the previous day. The dataset description

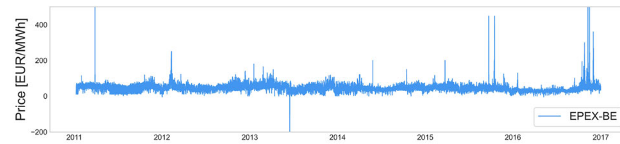


(a) Electricity price curve of NP dataset.

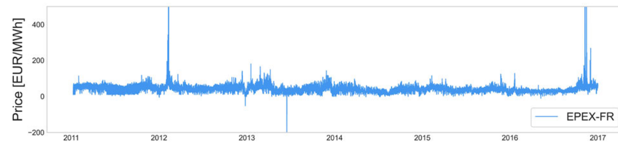


(b) Electricity price curve of PJM dataset.

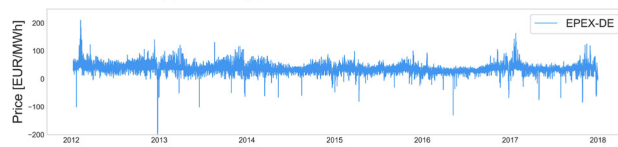
Figure 3. Visualization of electricity price characteristic curves in NP and PJM datasets. (a) Electricity price curve of NP dataset. (b) Electricity price curve of PJM dataset.



(a) Electricity price curve of EPEX-BE dataset.



(b) Electricity price curve of EPEX-FR dataset.



(c) Electricity price curve of EPEX-DE dataset.

Figure 4. Visualization of electricity price characteristic curves in EPEX-BE, EPEX-FR and EPEX-DE datasets. (a) Electricity price curve of EPEX-BE dataset. (b) Electricity price curve of EPEX-FR dataset. (c) Electricity price curve of EPEX-DE dataset.

suggests that these two types of indicators have the most significant impact on forecasting. The time period covered is from January 9, 2011, to December 31, 2016.

EPEX-FR (Electricity Exchange France): This dataset represents the electricity market in France. It contains day-ahead load forecasts and day-ahead wind power generation forecasts for the previous day. The time span matches that of the BE dataset, extending until December 31, 2016.

EPEX-DE (Electricity Exchange Germany): This dataset covers the electricity market in Germany and spans from January 9, 2012, to December 31, 2017. It combines solar load forecasts and wind power generation forecasts for the TSO Amprion area.The visualizations of above three datasets are presented in Figure 4.

3.2. Experiment setting

In the context of the aforementioned five datasets, we conducted comparisons with six of the most prevalent models in current research: Informer (H. Zhou et al., 2021), FEDformer (T. Zhou et al., 2022), Reformer (Kitaev et al., 2020), Transformer (Li et al., 2019), DLinear (Zeng et al., 2023), and Crossformer (Y. Zhang & Yan, 2022). For each dataset, we performed predictions with input time series lengths of 96, 128, 192, 256, all with a prediction length of 96. The training dataset comprised data from the first 4 years and 8 months, totaling 40,034 time sequences, while the testing dataset included data from the subsequent 1 year and 4 months, with a time sequence length of 11,425.

In our experimental protocol, we tailor the architecture to the specific model. For MILET, we specify a single encoder layer, while for all other models, we integrate a double-layer encoder design. The hidden state dimension is configurable from 16 to 512. We select the ADAM optimizer with a learning rate varying from 1e−4 to 1e−2. The batch size is fixed at 32, and the training duration is set to 10 epochs, with an early stopping criterion of patience 3. The initial weight for the linear model wlin is randomly drawn between 0.5 and 1. All experiments are conducted on a single GPU, a NVIDIA GeForce RTX 4060 with 8GB of memory, ensuring that our setup is both efficient and practical.

3.3. Experiment results

In the VMD-SE phase, the first step is to determine the value of K, which corresponds to the number of decompositions performed by the VMD algorithm. The choice

Table 1. Decomposition centre frequency for different K values in the EPEX-DE dataset.

K	Decomposition centre frequency									
$K = 4$	1.21×10^{-4}	0.083	0.247	0.374						
$K = 5$	1.21×10^{-4}	0.083	0.206	0.290	0.354					
$K = 6$	1.20×10^{-4}	0.083	0.165	0.248	0.333	0.405				
$K = 7$	1.20×10^{-4}	0.083	0.124	0.206	0.290	0.333	0.416			
$K = 8$	1.08×10^{-4}	0.040	0.083	0.166	0.247	0.291	0.374	0.458		
$K = 9$	1.07×10^{-4}	0.040	0.083	0.165	0.208	0.290	0.333	0.357	0.459	
$K = 10$	1.07×10^{-4}	0.041	0.083	0.165	0.207	0.249	0.291	0.333	0.375	0.459

Table 2. Sample entropy of each IMF when $K = 8$.

IMF	Sample entropy
Original series	2.803
1	0.561
2	0.687
3	0.632
4	0.408
5	0.169
6	0.183
7	0.373
8	0.333

of K is critical as it affects the similarity between the generated IMFs and the completeness of information extraction. Each mode produced by VMD has a corresponding central frequency.

If K is chosen too small, it may lead to incomplete information extraction. Conversely, if K is too large, it can result in increased redundancy and noise. To determine the optimal value of K, we perform VMD with different K values and monitor the central frequency of the final modal. When the central frequency of the last modal stabilizes and no longer changes significantly, it indicates the optimal K value for the decomposition. Without loss generality, we selected the EPEX-DE dataset and computed its decomposition results for different K values to illustrate this process. The centre frequencies for different K are provided in Table 1.

Once the value of K is determined, we obtain a series of IMFs, which exhibits distinct nonlinear characteristics and oscillatory patterns. However, they also maintain a certain level of similarity among them. To alleviate the network's computational burden and enhance efficiency, we use SE to assess the information content of these IMFs, thereby identifying significant differences among them, and simplifying their representation. Table 2 is the SE calculations for the various modals when $K = 8$ for the EPEX-DE datasets:

We observed that the entropy value of the original sequence is relatively high, surpassing the entropy values of all the decomposed sequences. This indicates that the original sequence exhibits the highest degree of information variability and is the most complex among them all. Among the 8 IMFs, we can distinctly observe a hierarchical distribution of entropy values. Specifically, IMF1, IMF2, and IMF3 have entropy values ranging between 0.55 and 0.70, while IMF4, IMF7, and IMF8 between 0.30 and 0.41. The remaining IMFs are distributed within the range of 0.16 to 0.20. The hierarchical distribution of entropy values numerically reflects the complexity characteristics of the corresponding sequences. Based on this, we classify and integrate the IMFs, ultimately synthesizing them into three distinct modes.

From the Table 3, it is evident that MILET consistently achieved the lowest error across the five datasets, regardless of input length. This reflects the model's generalization capability and validates the excellent performance of multidimensional and single-dimensional linear integration in the context of electricity price datasets. It can prove that MILET improves the performance and accuracy of EPF.

Furthermore, we have gathered data on the number of parameters and the duration of training for each model under various input lengths. It is crucial to recognize that the time required for training is not exclusively dictated by the GPU; there are also tasks that necessitate CPU engagement for data integration and preservation throughout the training phase. A scatter plot depicting the training duration for model parameters can be found in Figure 5.

3.3.1. Ablation experiments 1

To validate the necessity of each component within the model for reducing error and assess whether single-dimensional feature extraction and cross-dimensional information fusion can effectively enhance model expressiveness or not, we conducted ablation experiments to verify the reliability of each component. We designed six different decomposed validation scenarios, as follows:

(1) Model 1: Retaining only the dual-layer OTSA-LSTM for the dataset in MIE.
(2) Model 2: Adding the RevIn operation on top of the scenario in Model 1.
(3) Model 3: Incorporating the three modal features into the scenario in Model 2.
(4) Model 4: Retaining only the processing layers for the three modals in MIE.
(5) Model 5: Retaining only the three modals and the linear layer.
(6) Model 6: Adding the RevIn operation on top of the scenario in Model 5.

The results are showned in Table 4. We can easily observe that the performance of MILET is relatively superior, while the overall performance of Model 6 is subpar. Comparing Model 1 with 2, as well as Model 5 and 6, we can infer that the inclusion of the RevIn layer leads to a reduction in prediction error. Additionally, the comparison between Model 4 and 5 indicates that adding a linear layer significantly enhances EPF accuracy. After the incorporation of a linear layer, the model's performance surpasses that of Crossformer, particularly excelling in capturing trend information.

The disparity between Model 6 and MILET can be attributed to MILET's reliance on multidimensional

Table 3. Comparison of current state-of-the-art models across five datasets.

Models		Informer		FEDformer		Reformer		Transformer		DLinear		Crossformer		MILET	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
	96	0.707	0.571	0.438	0.444	0.615	0.563	0.489	0.480	0.380	0.378	0.370	0.385	**0.272**	**0.303**
	128	0.557	0.509	0.421	0.425	0.746	0.635	0.495	0.502	0.367	0.369	0.355	0.375	**0.283**	**0.312**
	192	0.558	0.510	0.421	0.426	0.746	0.636	0.496	0.503	0.367	0.370	0.367	0.384	**0.298**	**0.328**
NP	256	0.571	0.531	0.431	0.432	0.641	0.567	0.423	0.451	0.363	0.368	0.321	0.353	**0.304**	**0.331**
	96	0.235	0.343	0.184	0.296	0.186	0.298	0.155	0.246	0.151	0.260	0.255	0.260	**0.131**	**0.226**
	128	0.228	0.327	0.184	0.293	0.183	0.261	0.156	0.245	0.147	0.252	0.197	0.289	**0.133**	**0.223**
	192	0.231	0.336	0.182	0.292	0.204	0.283	0.200	0.248	0.145	0.246	0.237	0.261	**0.142**	**0.224**
PJM	256	0.355	0.376	0.178	0.286	0.173	0.269	0.162	0.248	0.144	0.243	0.179	0.245	**0.135**	**0.220**
	96	0.966	0.564	0.878	0.448	0.848	0.479	0.848	0.475	0.823	0.423	0.743	0.406	**0.700**	**0.328**
	128	0.859	0.481	0.819	0.423	0.823	0.465	0.762	0.394	0.794	0.387	0.739	0.386	**0.696**	**0.323**
	192	0.857	0.472	0.803	0.413	0.847	0.481	0.823	0.452	0.792	0.380	0.714	0.368	**0.683**	**0.325**
EPEX-BE	256	0.887	0.488	0.806	0.420	0.834	0.463	0.792	0.430	0.790	0.383	0.718	0.359	**0.670**	**0.322**
	96	0.517	0.383	0.453	0.329	0.503	0.367	0.913	0.388	0.459	0.329	0.455	0.306	**0.400**	**0.238**
	128	0.602	0.406	0.459	0.329	0.490	0.367	0.585	0.380	0.438	0.312	0.398	0.252	**0.381**	**0.232**
	192	0.522	0.378	0.452	0.319	0.484	0.349	0.959	0.385	0.418	0.291	**0.394**	0.246	0.499	**0.237**
EPEX-FR	256	0.514	0.367	0.479	0.341	0.516	0.373	0.556	0.350	0.406	0.278	0.409	0.271	**0.375**	**0.231**
	96	0.672	0.556	0.712	0.572	0.628	0.532	0.657	0.527	0.679	0.515	0.610	0.498	**0.486**	**0.432**
	128	0.674	0.553	0.704	0.573	0.645	0.535	0.637	0.526	0.630	0.494	0.579	0.491	**0.493**	**0.440**
	192	0.807	0.598	0.700	0.572	0.632	0.534	0.682	0.541	0.613	0.489	0.570	0.485	**0.516**	**0.446**
EPEX-DE	256	0.802	0.605	0.705	0.579	0.666	0.552	0.648	0.515	0.606	0.486	0.595	0.489	**0.556**	**0.465**

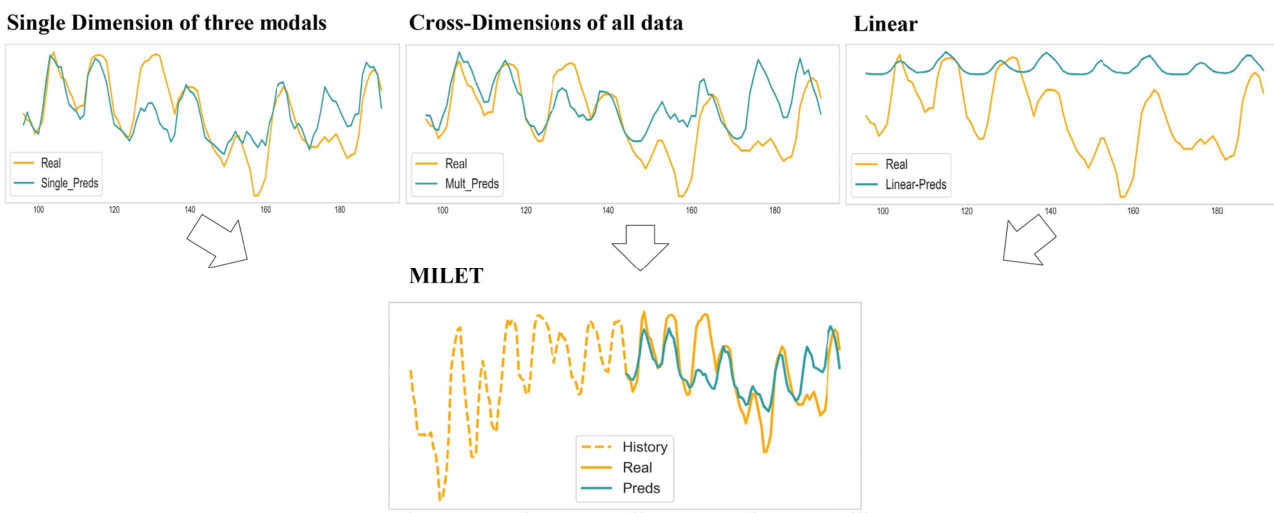


Figure 5. The training time and parameter number statistics of each model under different input lengths.

Table 4. Ablation experiment 1: MILET performance comparison for six component models.

Models		1		2		3		4		5		6		Crossformer		MILET	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
	96	0.646	0.515	0.607	0.492	0.677	0.544	0.778	0.565	0.516	0.448	0.529	0.451	0.610	0.498	**0.486**	**0.432**
	128	0.630	0.501	0.604	0.490	0.512	0.447	0.554	0.449	0.529	0.449	0.513	0.442	0.579	0.491	**0.493**	**0.440**
	192	0.659	0.525	0.621	0.496	0.516	0.450	0.564	0.457	0.541	0.461	0.521	**0.445**	0.570	0.485	**0.516**	0.446
EPEX-DE	256	0.647	0.517	0.655	0.519	0.581	0.478	0.576	0.469	**0.536**	**0.459**	0.538	0.460	0.595	0.489	0.556	0.465

information. Considered with Models 2, 6, and MILET, it is evident that there is indeed a distinction between single-dimensional and multidimensional information, and they can be effectively fused, contributing positively to the enhancement of EPF accuracy. We summarize the above ablation experiments into an integration of three prediction modes, as shown in Figure 6.

From the Figure 6, we can observe that the integration of three modal information and cross-dimensionality yields distinct results, each displaying variations in trends

Table 5. Ablation experiment 2: comparison of MILET to remove a channel respectively.

Models		Channel 1		Channel 2		Channel 3		Channel 4		Channel 5		MILET	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
EPEX-DE	96	0.499	0.435	0.506,	0.438	0.502	0.435	0.491	0.433	0.532	0.450	**0.486**	**0.432**
	128	0.521	0.451	0.516	0.447	0.500	0.433	0.494	**0.434**	0.521	0.445	**0.493**	0.440
	192	0.531	0.453	0.531	0.453	0.527	0.448	0.516	0.449	0.539	0.456	**0.516**	**0.446**
	256	0.576	0.472	0.576	0.477	0.566	0.467	**0.530**	**0.451**	0.575	0.478	0.556	0.465

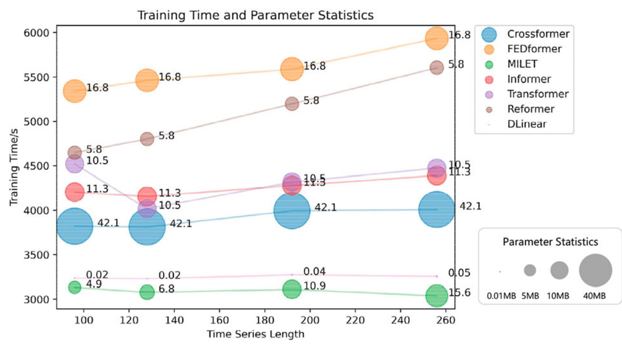


Figure 6. Single-dimensional prediction and multi-dimensional crossover prediction models relying on pattern decomposition are combined with linear trend prediction models for prediction visualization.

and peaks for EPF. The linear component, primarily capturing trends in relevant factors, exhibits only slight fluctuations in peaks.

3.3.2. Ablation experiment 2

We need to elucidate the necessity of the five MIE channels. To determine the necessity of the five channels, this study systematically removes one of the channels at a time. It then compares the performance of the model without the specific channel with the complete MILET model to assess whether it results in a reduction in prediction error or not.For the aforementioned experiments, we conducted comparisons using the EPEX-DE dataset. The experimental results are presented in the Table 5.

In the Table 5, the first five models, in sequential order, represent the removal of specific channels: high regularity mode, medium regularity mode, low regularity mode, other features, and all features combined. From the table, we can observe that removing any one of these channels has an impact on the model's prediction accuracy. In most cases, the MILET performs optimally. Particularly in the case of channel 4, which involves the removal of other features, we observe that this model generally achieves the second-best performance and even attains the best performance when the input length is 256. Although other features contribute some information to the EPF, the degree of correlation between the two types of features varies. Longer sequences of other inputs features bring more relevant information for EPF while introduce more irrelevant information.

4. Discussion and conclusion

We have introduced a Transformer-based model called MILET, which combines the extraction of single-dimensional features and cross-dimensional feature dependencies in short-term EPF after modal decomposition. We propose OTSA, based on TSA from Crossformer, but without the need for dimension splitting. OTSA directly processes all the input information without loss. OTSA can be applied to extract single-dimensional information. Using the VMD-SE method to obtain modals with different complexities and information content, we capture information within a single dimension and across dimensions using the OTSA-LSTM-based MIE. Overmore, our model can capture trend information for other auxiliary features using linear layers and linearly combine all the obtained information. Experimental results on five electricity price datasets demonstrate that MILET outperforms previous state-of-the-art techniques.

In EPF, it's crucial to consider various factors, including the impact of other relevant factors. We're exploring the inclusion of trading market and temperature-related features to enhance predictions. Transitioning from point to interval forecasting can improve reliability, which is significant for practical revenue assessments. Future research may integrate quantile regression into advanced models to improve accuracy.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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